Sparsh-Vaani

Raspberry Pi Library Installation & Commands Guide

Project scope

For the Multi-Sensory Medicine Label Reader and optional 6-compartment smart pill dispenser. This guide covers OCR, camera/image processing, voice output, USB serial communication with ESP32, and optional IoT/web components.

1. Update Raspberry Pi OS

```
sudo apt update
sudo apt upgrade -y
```

2. Install system packages

```
sudo apt install -y python3-pip python3-venv python3-opencv \
  python3-numpy python3-pil tesseract-ocr tesseract-ocr-eng \
  espeak-ng
```

Optional: for the official Raspberry Pi Camera Module on Raspberry Pi OS Bookworm, install Picamera2 with `sudo apt install -y python3-picamera2`. For a USB webcam, Picamera2 is usually not required.

3. Create a project folder and virtual environment

```
mkdir -p ~/Sparsh-Vaani
cd ~/Sparsh-Vaani
python3 -m venv --system-site-packages .venv
source .venv/bin/activate
```

4. Install Python packages

```
python -m pip install --upgrade pip
python -m pip install pytesseract pyserial requests smbus2 gpiozero flask
```

OpenCV, NumPy and Pillow are installed from Raspberry Pi OS packages above. The `--system-site-packages` option lets the virtual environment access those packages.

5. Create requirements.txt for GitHub

```
cd ~/Sparsh-Vaani
nano requirements.txt
```

Paste these lines into the file:

```
pytesseract
pyserial
requests
smbus2
gpiozero
flask
```

Save in nano: press Ctrl+O, press Enter, then Ctrl+X. Reinstall later with:

```
source ~/Sparsh-Vaani/.venv/bin/activate
python -m pip install -r ~/Sparsh-Vaani/requirements.txt
```

6. Test the installation

```
python -m pip list
tesseract --version
python -c "import cv2, pytesseract; print('OCR libraries working')"
python -c "import serial; print('pyserial working')"
espeak-ng "Sparsh Vaani system is ready"
```

7. Export installed Python package versions

```
cd ~/Sparsh-Vaani
python -m pip freeze > requirements-full.txt
```

This records Python packages in the active environment; it does not record apt-installed system software such as Tesseract.

8. Library purpose reference

Package / software	Purpose
Tesseract + pytesseract	Extract printed text from medicine labels
OpenCV (cv2) + NumPy	Camera image processing
Pillow	Image loading and manipulation
pyserial	USB serial communication between Raspberry Pi and ESP32
requests	HTTP requests to IoT services
smbus2	I²C access for compatible peripherals
gpiozero	Raspberry Pi GPIO control
Flask (optional)	Local web interface
espeak-ng	Text-to-speech voice output
Picamera2 (optional)	Official Raspberry Pi Camera Module support

Important hardware notes

• MAX98357A is an I²S amplifier; installing Python libraries alone does not enable/configure I²S audio.
• Use a suitable external regulated supply for servos and connect its ground to ESP32 ground. Do not power servos from 3.3V.
• An IR sensor can detect a tablet passing the chute, but cannot prove swallowing. Use an explicit patient confirmation button or voice prompt.
• IoT alerts need Wi-Fi/network access and configuration of the chosen service (for example, Blynk and email).